ERROR DETECTING AND CORRECTING CODE

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ERROR DETECTING AND CORRECTING CODE

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ERROR DETECTING AND CORRECTING CODE

ERROR DETECTING AND CORRECTING CODE:

- Whenever information is transmitted from one device (transmitter) to another device (receiver), there is possibility that errors can occur. So the receiver does not receive the identical information.
- This means either 0 changes to 1 or 1 changes to 0.
- To maintain the data integrity between transmitter and receiver, extra bit or more than one bit are added in the transmitted data.
- The data along with the extra bit/bits forms the *Code*.
- Codes which allow only error detection are called *Error Detecting Codes*.
- Codes which allow error detection and correction are called *Error Detecting And Correcting Code*.
- There are some methods for detection. Most commonly used scheme for error detection is *Parity Method*.

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PARITY BIT METHOD:

- A parity bit is an extra bit that is attached to a group that is being performed from one location to another.
- The parity bit is made either 0 or 1, depending upon the number of 1's that are contained in the core group.
- There are two methods (i) Even parity method (ii) Odd parity method.

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EVEN PARITY METHOD: In this method the value of the parity bit is chosen so that the total number of 1s in the code group including parity bit is an even number. For example, the group is 1000011. This is ASCII is "C". In this group the number of 1s is three. Hence we will add 1 to make the total 1s is an even number.

$\boxed{1}$ 1 0 0 0 0 1 1

Added Parity Bit

Another example, 1000001, this is ASCII is "A". Hence

$\boxed{0}$ 1 0 0 0 0 0 1

Added Parity Bit

The parity bit can be placed at either end of the code group but is usually placed to the left of the MSB.

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ODD PARITY METHOD: In odd parity method, the parity bit chosen so that the total number of 1s in the code group is an odd number. For example

For code group 1000001, the assigned parity bit would be 1, i.e. 11000001

For code group 1000011, the assigned parity bit would be 0, i.e. 01000011

EXAMPLE 1: A transmitter is sending ASCII coded data to receiver with an even parity bit. Show the actual codes when the transmitter is sending the message "HELLO".

SOLUTION:

	ASCII CODE
H	0 1 0 0 1 0 0 0
E	1 1 0 0 0 1 0 1
L	1 1 0 0 1 1 0 0
L	1 1 0 0 1 1 0 0
O	1 1 0 0 1 1 1 1

The first bit in red color is the attached even parity bit

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EXAMPLE 2: Attach an odd parity bit to the ASCII code for ‘Y’ and express the result in hexadecimal.

SOLUTION: ASCII Code for ‘Y’ = 101 1001

Adding odd parity, the new code will be 1 0 1 1 0 0 1

$$1\ 0\ 1\ 1\ 0\ 0\ 1 = 59_{16}$$

EXAMPLE 3: Why can't the parity bit method detect a double error in transmitted data?

SOLUTION: Parity bit method would not work if two bits were in error, because two errors would not change the “ODDNESS” or “EVENNESS” of the number of 1s in the code.

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EXAMPLE 4: The received code is 10000001. Check whether code is correctly received or not if odd parity is used.

ANSWER: The received code has even parity, hence the code is not received correctly.

EXAMPLE 5: Attach an even parity bit to the BCD code for decimal 69.

ANSWER:

Decimal Number: 6 9

BCD Code: 0 1 1 0 1 0 0 1

After adding parity bit: 0 0 1 1 0 1 0 0 1

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HAMMING CODE:

- Hamming code also known as error detecting and error correcting code.
- Hamming code detects the bit error and also identify which bit is in error. So that it can be corrected.
- Hamming code uses the concept of *parity* and *parity bit*.
- Computing parity involves counting the number of ones in a unit data, and adding either a zero or a one to make the count odd (for odd parity) or even (for even parity).
- To calculate even parity, the XOR operator is used.
- To calculate odd parity the XNOR operator is used.

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NUMBER OF PARITY BITS: Number of parity bits depend on the number of information bits. If the number of information bit is ‘x’, then number of parity bits q is determined by (trial and error method)

$$2^q \geq x + q + 1$$

POSITION OF PARITY BITS IN THE CODE: the parity bits are located in the position that are numbered corresponding to ascending powers of two (1, 2,4, 8, 16, …..). There for seven bit code, locations of parity bits and information bits are

m_7	m_6	m_5	P_4	m_3	P_2	P_1

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PROCEDURE FOR CALCULATING THE HAMMING CODE:

(i) Mark all bit positions that are power of two as parity bits. (position 1,2,4,8,16,32,64,…)

(ii) All other bit positions are for the data to be encoded. (positions 3,5,6,7,9,10,11,12…)

(iii) Each parity bit calculates the parity for some of the bits in the code word. The position of parity bits determines the sequence of bits that it alternately checks and skips.

Position 1: check 1 bit, skip 1 bit, check 1 bit, skip 1 bit etc. (1, 3, 5,7, 9, …...)

Position 2: check 2 bit, skip 2 bits, check 2 bits etc (2,3,6,7,10,11,14,15……)

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Position 4: check 4 bits, skip 4 bits, check 4 bits etc. (4,5,6,7,12,13,14,15….)

Position 8: check 8 bits, skip 8 bits etc. and so on.

(iv) Set a parity bit 1 if the total number of 1 in the positions it checks is odd.

Set a parity bit 0 if total number of 1 in the positions it checks is even.

EXAMPLE 1: Encode the binary word 1011 into seven bit even parity Hamming code. (I.U. 2018-19)

SOLUTION: Find the number of parity bits by using equation (1).

$q = 0,\ 2^0 >$	$4 + 0 + 1 = 5$	False
$q = 1,\ 2^1 >$	$4 + 1 + 1 = 6$	False
$q = 3,\ 2^3 >$	$4 + 3 + 1 = 8$	True

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$\therefore$ q = 3, three parity bits are sufficient.

Number of boxes = x + q = 4 +3 =7

m_7	m_6	m_5	P_4	m_3	P_2	P_1
1	0	1		1		

Now check for P_1. skip 1 bit. Bit location 3,5 and 7 have three ones, therefore to have even parity P_1 must be 1. $P_1 = 1$

m_7	m_6	m_5	P_4	m_3	P_2	P_1
1	0	1		1		1

Now check for P_2. skip 2 bit. Bit location 3,6 and 7 have two 1, therefore to have even parity P_2 must be 0. $P_2 = 0$

m_7	m_6	m_5	P_4	m_3	P_2	P_1
1	0	1		1	0	1

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Now consider 4 bits (boxes) from P_4 and skip next four if any.

Bit location 5,6,7 have two 1, therefore to have an even parity P_4 must be 0.

m_7	m_6	m_5	P_4	m_3	P_2	P_1
1	0	1	0	1	0	1

m_7	m_6	m_5	P_4	m_3	P_2	P_1
1	0	1	0	1	0	1

$\therefore$ Seven bit Hamming Code = 1010101

EXAMPLE 2: Encode $(10110011)_2$ into seven bit even parity Hamming Code.

SOLUTION:

$q = 1, \; 2^1 \geq 8 + 1 + 1 = 10$ False

$q = 2, \; 2^2 \geq 8 + 2 + 1 = 11$ False

$q = 4, \; 2^4 \geq 8 + 4 + 1 = 13$ True

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No. of Boxes = x + q = 8 + 4 = 12

m_{12}	m_{11}	m_{10}	m_9	P_8	m_7	m_6	m_5	P_4	m_3	P_2	P_1
1	0	1	1		0	0	1		1		

For P_1: Bit locations 3,5,7,9 and 11 have three ones, therefore to have an even parity, P_1 must be 1

m_{12}	m_{11}	m_{10}	m_9	P_8	m_7	m_6	m_5	P_4	m_3	P_2	P_1
1	0	1	1		0	0	1		1		1

For P_2: Bit locations 3,6,7,10 and 11 have two ones, therefore to have an even parity, P_2 must be 0

m_{12}	m_{11}	m_{10}	m_9	P_8	m_7	m_6	m_5	P_4	m_3	P_2	P_1
1	0	1	1		0	0	1		1	0	1

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For P_4: bit locations 5,6,7,12 have two ones, therefore to have even parity P_4 must have 0.

m_{12}	m_{11}	m_{10}	m_9	P_8	m_7	m_6	m_5	P_4	m_3	P_2	P_1
1	0	1	1		0	0	1	0	1	0	1

For P_8: Consider 8 boxes, skip next 8 boxes. Bit location 9,10,11, 12 have three ones, for even parity P_8 must have 1.

m_{12}	m_{11}	m_{10}	m_9	P_8	m_7	m_6	m_5	P_4	m_3	P_2	P_1
1	0	1	1		0	0	1	0	1	0	1

$\therefore$ Hamming Code = 101110010101

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DETECTING AND CORRECTING AN ERROR:

After parity check, formed the binary word. This word gives the bit location. Where error has occurred. If word has all bit 0 then there is no error in Hamming code.

Note: Take the resulting bit P_1 as LSB.

EXAMPLE 3: Assume that the even parity Hamming code 0110011 is transmitted and that 0100011 is received. The receiver does not know what was transmitted. Determine bit location where error has occurred using receiving code.

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SOLUTION:

m_7	m_6	m_5	P_4	m_3	P_2	P_1
0	1	0	0	0	1	1

For P_1: Bit location 1,3,5,7 have one 1. parity checks for even parity is wrong. i.e. 1

For P_2: Bit location 2,3,6,7 has two 1. parity checks for even parity is correct. i.e. 0

For P_3: Bit location 4,5,6,7 has one 1. parity checks for even parity is wrong. i.e. 1

Resultant word (from P_3 to P_1) = 101. This means the bit in the number 5 location is in error. It is 0 and must be 1. Therefore, correct code is 0110011, which match as with the correct code.

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EXAMPLE 4: The Hamming code 101101101 is received. Correct it, if any errors. There are four parity bits and odd parity is used.

SOLUTION:

m_9	P_8	m_7	m_6	m_5	P_4	m_3	P_2	P_1
1	0	1	1	0	1	1	0	1

For P_1: P_1 checks location 1,3,5,7 and 9

No. of ones = 4

Parity check for odd parity is wrong i.e. 1 (LSB)

For P_2: P_2 checks location 2,3,6 and 7

No. of ones = 3

Parity check for odd parity is correct i.e. 0

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For P_4: P_4 checks location 4,5,6 and 7

No. of ones = 3

Parity check for odd parity is correct i.e. 0

For P_8: P_8 checks location 8 and 9

No. of ones = 1

Parity check for odd parity is correct i.e. 0 (MSB)

Resultant word = 0001

It means that the bit in number 1 location is in error. It is one (1) it must be 0.

$\therefore$ Correct code = 101101100

THANK YOU

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